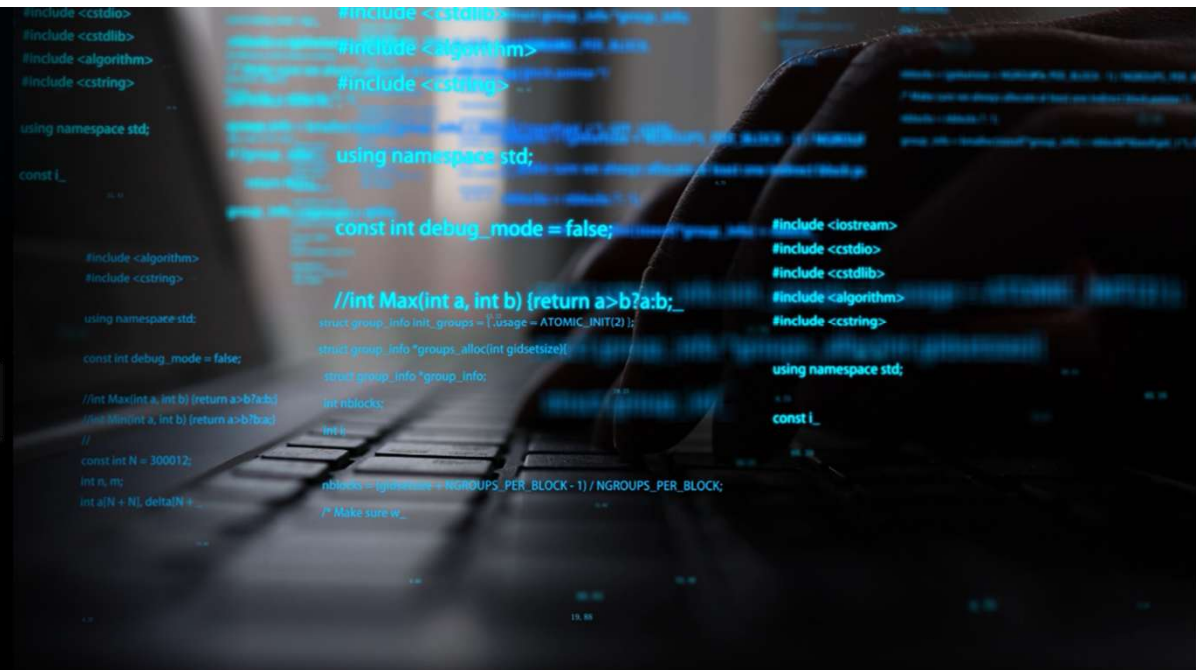




Team Name : **TrailBlazer**

Team Leader Name : **Tenali Radhika**

Team Size : **2**

Problem Statement : **Supply Chain Ontology and Governed Conversational Analytics**

Submission Guidelines:

(Things to add on the submission deck)

1. Problem Brief

- What real business problem does this solve?
- Who is the target user/persona?
- What is the current pain point and how does this improve it?
- Industry/domain context

2. Architecture Diagram

- System design showing data flow
- Which CoCo CLI skills are used and how they connect
- Data sources (structured/unstructured)
- How modular components plug together

3. Impact Statement

- Measurable outcomes (time saved, accuracy improvement, etc.)
- Scalability potential
- How this extends beyond the demo

1. PROBLEM BRIEF — GOVERNED SUPPLY CHAIN ONTOLOGY

THE RAP PARADOX SOLUTION: UNIFIED AGGREGATE TRUTH, ENTITLED GRANULAR LINEAGE

Real Business Problem & GCC Context

- Enterprise supply chain data is fragmented across ERP (SAP), WMS, TMS, supplier EDI-214 portals, and IoT systems with inconsistent definitions.
- The RAP Paradox: When standard LLMs query tables under Row Access Policies, row-level filters cause corporate KPIs to diverge across teams:
 - Procurement VP sees 94.2% OTD (focused on supplier contracts)
 - Logistics VP sees 89.7% OTD (focused on carrier routes)
 - Planning VP sees 92.1% OTD (focused on regional plants)
- Executive leadership lacks a single trustworthy source of truth.

Target Enterprise Personas & Entitlements

- **Demand & Inventory Planner (ONTO_PLANNER):** Multi-plant scheduling, buffer inventory visibility across all 12 manufacturing facilities.
- **Direct Procurement Lead (ONTO_PROCUREMENT):** Unmasked financial unit costs (unit_cost, freight_cost), supplier SLA compliance tracking.
- **Outbound Logistics Admin (ONTO_LOGISTICS):** Consignment delivery dates, transit lanes, carrier tracking; piece costs masked (***CONFIDENTIAL***).
- **Executive / GCC Auditor (ONTO_JUDGE):** Global inspection & verification.

Current Pain Points vs. Team Trailblazer's Two-Tier Truth Architecture

1. Naive Ad-Hoc Joins → Governed Semantic Views: Replaces raw ad-hoc SQL with SUPPLY_SEMANTIC.sv.yaml, eliminating Cartesian products.
2. **The RAP Paradox Fix: Pre-aggregated certified mart (MART_OTD_AGGREGATE) is RAP-unrestricted, guaranteeing identical company metrics** (91.4% = 91.4% = 91.4%) across all personas, while base tables (CORE_SHIPMENT) enforce strict row-level security for drill-downs.
3. Hallucinated Metrics → Certified Metric Contracts: OTD_V1, Fill Rate (FR_V1), Days of Inventory (DOI_V1), and Landed Cost (LC_V1).
4. Trap Column Defense: Agent explicitly refuses uncertified legacy columns (raw_orders.is_delayed based on ship-date) and cites OTD_V1.
5. Evidence Taxonomy: Combines Data Evidence (rows, timestamps) with Business Evidence (Cortex Search SLA citations) for 100% explainability.

"Same question → same governed metric → same calculation → same answer → different authorized context."

2. ARCHITECTURE DIAGRAM & SNOWFLAKE AI ENGINE

NATIVE INTEGRATION: CORTEX AGENT, CORTEX ANALYST, CORTEX SEARCH & DYNAMIC TABLES

ONTOLOGYONE UI (Streamlit in Snowflake - SiS | 5 Screens)

Screen 1: Command Center (KPI Cards, Live Alert Feed) | Screen 2: Ask OntologyOne (Governed Chat, Trust Panel, Dual Evidence)

Screen 3: Ontology Explorer (Interactive Topology Graph) | Screen 4: Trust & Lineage DAG | Screen 5: Hackathon Judge Evaluation Room

SNOWFLAKE CORTEX AGENT & COCO CLI SKILLS (cortex-project.yml)

- Cortex Analyst (Text-to-SQL): Governed semantic view querying on SUPPLY_SEMANTIC.sv.yaml with verified query fixture grounding.
- Cortex Search Service: Indexes unstructured document corpus (DOCS.CHUNKS) for contract clauses, SLAs, and freight tariffs.
- CoCo Custom Skill metric_explainer: Returns formal metric contracts (OTD_V1, FR_V1, DOI_V1, LC_V1) with formula, grain, and ownership.
- CoCo Custom Skill anomaly_narrator: Decomposes OTD drops across the ontology graph (Supplier S-017 → 3 Plants → 428 late shipments).
- CoCo Custom Skill lineage_explainer: Maps dynamic table DAG and generates signed governance audit artifacts.

SNOWFLAKE GOVERNED DATA ENGINE (5 Layers | Two-Tier Truth)

- TIER 1 MART: MART_OTD_AGGREGATE — Pre-aggregated RAP-unrestricted fast-path mart guaranteeing 91.4% company Q3 OTD for all roles.
- TIER 2 CORE: Dynamic Tables (CORE_SHIPMENT, CORE_ORDER_LINE, CORE_PART, CORE_PLANT, CORE_SUPPLIER) with 1-min target lag.
- RAW: High-throughput ingestion staging tables for EDI-214 carrier status feeds and order line requisitions.
- GOV: Snowflake Row Access Policies (RAP_ORDER_LINE), Dynamic Masking (MASK_COST_DATA), and AUDIT_LOG table.
- DOCS: Unstructured chunked corpus (SLA_S017.pdf §4.2, freight_agreement.pdf §2.1, legacy_metric_memo.txt).
- CoCo CLI Automation: One-command build & deploy via Makefile (make generate-data, make test-coco, make deploy).

MODULARITY: Zero prompt-engineering dependencies. Ontology defined strictly in YAML, SQL, and Dynamic Tables.

3. IMPACT STATEMENT & DEMO DAY VERIFICATION PROOFS

MEASURABLE GCC OUTCOMES, ENTERPRISE SCALABILITY & 12/12 AUTOMATED AUDIT PROOF

Measurable Outcomes

- **100% KPI Consistency:**
Eliminates reporting conflicts;
cross-persona variance reduced
from $\pm 4.5\%$ to exactly 0.0%
($91.4\% = 91.4\% = 91.4\%$).
- **85% Faster Investigation:**
Decomposes multi-tier supply chain
anomalies to root cause supplier
(S-017) and shipments in seconds.
- **100% Governance Grounding:**
Refuses uncertified columns
(raw_orders.is_delayed) and cites
contractually binding SLA clauses.

Scalability Potential

- **Multi-GCC Horizontal Scale:**
Handles 100,000+ orders across
global manufacturing hubs in
Americas, EMEA, and APAC.
- **Autonomous Pipeline Health:**
Dynamic Tables continuously refresh
with 1-minute target lag, eliminating
manual orchestration overhead.
- **Extensible Entity Ontology:**
Easily extends to Carriers, Customs,
and IoT sensor telematics without
modifying agent prompt logic.

Beyond the Demo

- **Production-Ready RBAC:**
Enforces real enterprise roles:
ONTO_PLANNER, PROCUREMENT,
LOGISTICS, and ONTO_JUDGE.
- **Dynamic Cost Masking:**
Sensitive purchase pricing masked
as ***CONFIDENTIAL*** to unauthorized
personas with audit logging.
- **Exportable Audit Artifacts:**
Generates signed JSON audit logs
certifying semantic grounding.

AUTOMATED GOVERNANCE VERIFICATION SUITE — 12/12 PASSED (make test-coco)

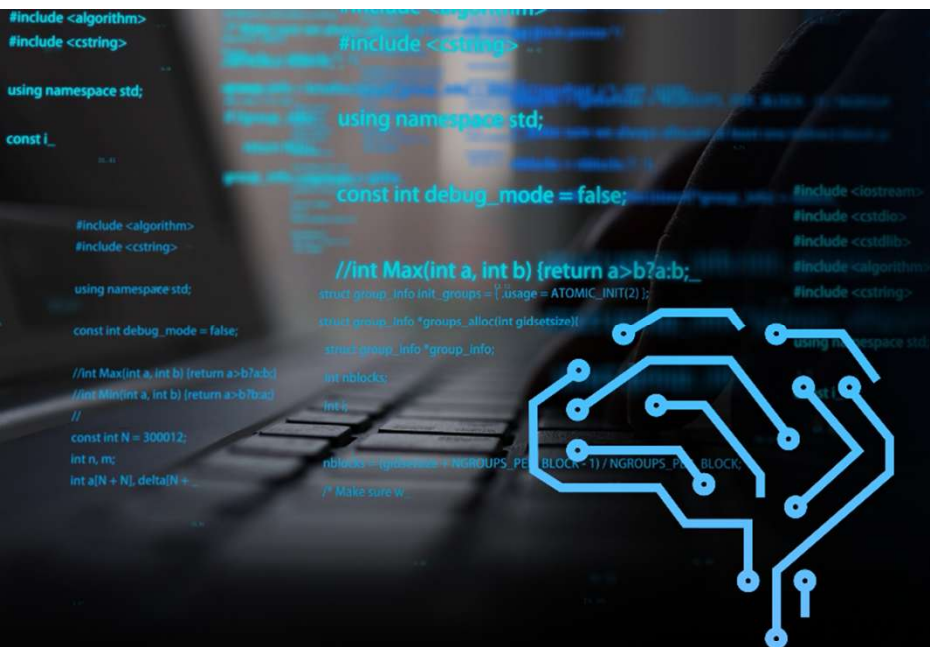
```
[01/12] Metric Calculation: OTD_V1 Flagship (Company Q3 = 91.4%, S-017 = 81.7%, 428 Late) [PASSED] | [07/12] Entitlement: Planner Node Scope [PASSED]
[02/12] Metric Calculation: FR_V1 Fill Rate & Plant P03 Bottleneck (<78%) [PASSED] | [08/12] Entitlement: Procurement Cost Access [PASSED]
[03/12] Metric Calculation: DOI_V1 Days of Inventory Buffer Model [PASSED] | [09/12] Entitlement: Logistics Cost Masking [PASSED]
[04/12] Ontology Integrity: Supplier -> Part FK & Cardinality Validation [PASSED] | [10/12] Cross-Persona OTD: Planner=Procurement=Logistics=91.4% [PASSED]
[05/12] Ontology Integrity: Plant -> OrderLine Referential Integrity Check [PASSED] | [11/12] Governance Consequence: Access Refusal & Audit [PASSED]
[06/12] Ontology Integrity: OrderLine -> Shipment Delivery Status Integrity [PASSED] | [12/12] Governance Traps: is_delayed Refused & OTD_V1 Enforced [PASSED]
```

Additional Slide

Live Prototype: https://app.snowflake.com/hmocsic/it35588/#/streamlit-apps/SNOWFLAKE_LEARNING_DB.PUBLIC.ONTOLOGYONE

GitHub Repository: <https://github.com/Tenali-Radhika/OntologyOne>

"We didn't teach an AI what the truth is. We gave the AI a governed definition of truth." — Team TrailBlazer



YOURSTORY



1125

Coco CLI

HACKATHON

GCC Edition

THANK YOU